MILITARY SPECIFICATION

YARN, COTTON

1. SCOPE

1.1 This specification covers yarn, cotton, carpet warp, suitable for hand weaving in occupational therapy.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-665	-	Boxes, Panerboard, Metal Stayed (including stay material)
PPP-B-676	-	Boxes, Set-up
PPP-T-45	-	Tape, Gummed, Paper Reinforced and Plain For Sealing and Securing

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8310

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Standard sample.- When a standard sample is available the yarn shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.2).

* 3.2 Yarn.- The yarn shall be made of clean cotton which has been carded, spun, and twisted into yarn. The yarn shall be carpet warp 8/4 ply ± 3 percent, 6 to 7-1/2 turns per inch of "S" twist in the ply. The plied yarn shall have a breaking strength of 7 pounds minimum.

3.3 Color.- The color of the yarn shall be natural, bleached white, or color as specified (see 6.1) and shall conform to an approved shade standard (see 6.3). When white is specified, the yarn shall be bleached using a peroxide type bleach. No tinting is permissible and the use of optical bleaches is not permitted. When natural is specified, the yarn shall be in natural shade.

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* 3.3.1 Labile sulfur.- The use of dyes and compounds containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished yarn shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.3.

3.3.2 Matching.- The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.3 Colorfastness.- The dyed and finished yarn shall show colorfastness to laundering equal to or better than the standard sample when tested as specified in 4.3. When no standard sample is established, or when the standard sample is not referenced for colorfastness, the dyed and finished yarn shall show fair fastness to laundering.

3.4 Put-up.- The yarn shall be put up on spools containing 840 yards ± 10 percent.

3.5 Fiber identification.- Each spool shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.6 Workmanship.- The finished yarn shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of end item.-

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* 4.2.2.1 Visual examination.- Defects found during this examination shall be classified in accordance with the defects listed in table I and regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. The sample unit for this examination shall be one spool. The acceptable quality level (AQL) for this examination shall be 2.5 major defects and 10.0 total defects per 100 units. The sample size shall be in accordance with inspection level I of MIL-STD-105. The lot size shall be expressed in units of one spool.

TABLE I. - Classification of defects

Examine	Defect	Classification Major	Minor
Cleanness	Spot or stain, clearly visible 1/		X
	Any burr, shive or foreign matter		X
Color	Wrong color	X	
	Off shade, i.e., not within established tolerances		X
Package	Cut, torn, chafed, or otherwise damaged package, affecting the free unhampered unwinding of yarn or affecting strength of yarn, or affecting the secure holding of yarn winds on the package.	X	
Type of package	Other than specified	X	

1/ At normal inspection distance (approximately 3 feet).

4.2.3 Examination for length.-

4.2.3.1 Examination for length per spool.- A defect with regard to length per individual spool shall be considered to exist if the yardage of any spool is not within the range specified in 3.4. The inspection level shall be S-3 and the acceptable quality level (AQL) shall be 4.0 defects per 100 units. For lots consisting of 500 or fewer units the sample size shall be ten and the acceptance number one. The lot size shall be the number of units in the inspection lot.

4.2.3.2 Average length.- A lot shall be unacceptable if the average length is less than the specified length. The minus tolerance of 10 percent shall not apply to the average length.

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4.3 Test methods.- The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit. The sample unit shall be one spool. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The lot size shall be expressed in units of one spool. The sample size (number of sample units) shall be in accordance with the following:

Lot size (spools)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. - Test methods

Characteristic	Requirement paragraph	Test method
Material (cotton)	3.2	1200 1/
Yarn size	3.2	4021
Twist direction	3.2	4050
Twist turns per inch	3.2	4054 2/
Yarn ply	3.2	Visual 3/
Breaking strength	3.2	4100
Labile sulfur	3.3.1	2020 1/
Colorfastness to laundering	3.3.3	5610

1/ A certificate of compliance shall be acceptable for this requirement.
2/ Twist contraction is not required.
3/ One determination per sample unit and results reported as "pass or fail".

4.4 Examination of preparation for delivery requirements.- An examination shall be made to determine that the packaging, packing and marking complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

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Examine	Defects
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping or inadequate stapling. Bulging or distortion of containers.
Packaging	Not as specified. Packages not as specified.
Packing	Not as specified. Number of packages not as specified. Mixed colors in same container. Number of spools per container not as specified. 1/

1/ For this defect one package shall be examined from each shipping container in the sample.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.- Twelve spools of yarn of one color only, put up as specified in 3.4, shall be packaged in a snug-fitting folding paperboard box conforming to variety 1, style III, type G, class j, group I of PPP-B-566; set-up paperboard box conforming to type II, variety 1, class A, style 4 of PPP-B-676; or metal stayed paperboard box conforming to style B of PPP-B-665. The lid for the boxes shall be not less than 1-inch in depth. Box closure shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade A of PPP-T-45

5.1.2 Level C.- The yarn shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A.- Ninety-six spools of yarn of one color only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636.

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5.2.2 Level B.- Ninety-six spools of yarn of one color only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SU) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.1), the shipping container shall be a grade V3c, V3s, V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification.

5.2.3 Level C.- Yarn, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate of such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color required (see 3.3).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.2 Standard sample.- For access to standard sample (see 3.1), address the procuring office issuing the invitation for bids.

6.3 The standard shade samples have been dyed by the use of the following dyes:

Black, 281	-	Direct Black 38, C.I. 30235
Blue, 282	-	Direct Blue 106, C.I. 51300
Brown, 283	-	Direct Brown 95, C.I. 30145
Gray, 284	-	Direct Black 3, C.I. 27710
Green, 285	-	Direct Green 26, C.I. 34045
Orange, 286	-	Direct Orange 34, C.I. 40215/20
Red, 287	-	Direct Red 80, C.I. 35780
Yellow, 288	-	Direct Yellow 28, C.I. 19555

6.4 Marginal identification.- The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications,

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corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of the document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

YARN, COTTON MIL-Y-[illegible](GL)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 65 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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